Extracorporeal membrane oxygenation for newborn respiratory failure: forty-five cases.

Almost all types of newborn respiratory failure are reversible. However, supportive treatment (oxygen and positive airway pressure) can damage the lung, and newborn respiratory failure remains a major cause of morbidity and death in infants. Prolonged extracorporeal membrane oxygenation (ECMO) provides life support while allowing the lung to "rest." We have used ECMO in 45 moribund newborn infants; 25 survived. Neonatologists referred patients who were unresponsive to maximal therapy. The right atrium and aortic arch were cannulated via the jugular vein and carotid artery. Heparin was infused continuously to main activated clotting time at 200 to 300 seconds. Airway oxygenation and pressure were reduced to low levels. Primary diagnoses were hyaline membrane disease, 14 (6 survived, 8 died); meconium aspiration, 22 (15 survived, 7 died); persistent fetal circulation including diaphragmatic hernia, 5 (3 survived, 2 died); and sepsis, 4 (1 survived, 3 died). Growth, development, and brain and lung function are normal in 20 of 25 survivors. ECMO decreased newborn respiratory failure mortality and morbidity rates in this phase I trial. A controlled randomized study is underway. The results suggest that ECMO may be effective in older patients if used before irreversible lung damage occurs.